Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/22/2025
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Detecting Pilot Spatial Disorientation to Trigger Active Countermeasures During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHP, OMNIBUS3 Human Research Program: Human Health & Performance Appendix E; Omnibus3-Appendix F
Start Date:	12/23/2022	**End Date:**	12/22/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/22/2026 per the PI (Ed., 10/29/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holder, Sherrie Ph.D. (Charles Stark Draper Laboratory Inc) Endsley, Tristan Ph.D. (Charles Stark Draper Laboratory Inc) Vance, Eric Ph.D. (University of Colorado, Boulder) Dixon, Jordan (Charles Stark Draper Laboratory, Inc.)		
Grant/Contract No.:	80NSSC23K0449		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	November 2024 Update Per the Principal Investigator: in coordination with the NASA Human Research Program, the research team has successfully modified its original proposal to more extensively and precisely define the countermeasure intervention that would be triggered by its spatial disorientation algorithm. See the Task Progress section for additional information. [Ed., 11/6/24.] During transit in microgravity, astronauts will reinterpret neurovestibular stimuli, prior to initial exposure to partial gravity when landing on the Moon or Mars. This poses a risk of spatial disorientation and impaired manual control performance during piloted planetary landings. Here, we propose to develop, validate, and assess a system for detecting when astronauts may become disoriented in real-time, such that it can be used to trigger active countermeasures for piloted planetary landings. Our approach leverages a well-validated computational model for human spatial orientation, now applied to partial gravity planetary landings. Incorporating microgravity neurovestibular adaptation, the vehicle motions of each landing trajectory are processed in real-time by the computational model to detect pilot spatial disorientation. We will assess the system using a ground-based lunar landing analog, combining a gravity transition (3 Gx) with a motion-based planetary landing simulation. First, we will experimentally tune and re-validate the computational model for detecting spatial disorientation, accounting for the effects of the recent gravity transition. Then, using the high-fidelity Disorientation Research Device, we will assess the benefit of the active countermeasure triggering system. Critically, this approach of triggering manual control countermeasures only when they are needed (i.e., when the pilot is about to be disoriented) avoids the added burden on the pilot to continuously process additional sensory information or otherwise have increased workload. We aim to deliver a validated performance support tool for triggering active countermeasures for pilot spatial disorientation during manually controlled lunar landings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing a means to estimate a human's perception of self-orientation and the potential for spatial disorientation in real-time, in order to trigger an intervention to enhance performance and safety. In addition to the application of piloted lunar landing, our approach can be applied to pilots of terrestrial aircraft or other vehicles in which spatial orientation perception is critical. Finally, our enhancement of models for human spatial orientation perception applies to humans more generally, such as high performance performance athletes (e.g., gymnasts), critical operators (e.g., scuba divers), or patient populations (e.g., mTBI patients of older adults at risk for impaired balance).
Task Progress:	In the third year of this project, we have been performing a series of human subject experiments. In the first experiment, we are revalidating our computational tool for the unique lunar landing-like motions, following an analog for a gravity transition, which we hypothesized will affect human spatial orientation perception. In a ground-based study, healthy subjects were first exposed to the Gx gravity transition analog using our laboratory's centrifuge, then they were exposed to various tilt (and in some cases translation) motions using the Tilt Translation Sled in our laboratory. Subjects were tasked with reporting spatial orientation perception during motions. Two cohorts have completed this study, with one cohort experiencing roll tilt and reporting perceptions of roll tilt and the second cohort experiencing pitch tilt and reporting perception of pitch tilt, each following the Gx gravity transition analog. We have used these datasets to refit our computational model for spatial orientation perception. In a second experiment, we systematically quantified the intensity of spatial disorientation, following various motion profiles. Following motions (which were designed to be more or less disorienting), subjects were suddenly presented with veridical instrument display information regarding their orientation and instructed to rate the severity/intensity of their spatial disorientation. Fourteen subjects completed the experiment of nominally 50 trials of motion and spatial disorientation rating. The results have been used to refine and fit parameters within our spatial disorientation algorithm to be able to predict the severity of spatial disorientation in real-time. Using a predictive modeling approach, we found quite accurate predictions of spatial disorientation using our computational model. In a final experiment, we will assess the efficacy of the spatial disorientation algorithm in triggering the adaptive display countermeasure during a piloted lunar landing. In two groups (control group and countermeasure group), subjects will pilot a motion-based lunar landing simulator. The control group will receive no countermeasure in order to quantify the decrements in piloting performance. In the countermeasure group, the validated algorithm for estimating spatial disorientation in real-time will trigger a countermeasure when the pilot is likely to be disoriented. This adaptive display will help the pilot become reoriented, only when it is necessary, without overburdening the pilot when they are not spatially disoriented. We will assess pilot performance, mental workload, and situation awareness. Initial testing will be performed in our laboratory in preparation for our capstone experiment in the Navy Aeromedical Research Laboratory – Dayton (NAMRU-D) Disorientation Research Device (DRD). In this third year of the project, substantial technical, scientific, and administrative progress has been made. This project has involved extensive coordination with colleagues associated with the NASA Human Research Program, NASA Langley, and NAMRU-D in coordination for our experiment using the DRD. Coordinating with colleagues at NASA Langley, we have successfully integrated the detailed implementation of our countermeasure adaptive display into their lunar landing simulation. We had a site visit to coordinate technical details of implementing our system (and other associated experimental needs) into their lunar-landing simulation and display environment. In addition to coordinating the countermeasure implementation, we have collaborated to design 21 unique, but similar, lunar landing scenarios (defined by initial altitudes and other parameters, followed by redesignations to specific landing sites) that will be used in our experiment at NAMRU-D. Finally, we have coordinated a data streaming protocol to enable the Langley lunar landing simulation to integrate with our system for detecting spatial disorientation in real-time. Finally, we have extensively coordinated with colleagues at NAMRU-D regarding plans for the capstone experiments using the DRD. This has included administrative coordination, including a signed CRADA document. We have written and gained approval from the local Institutional Review Board (IRB) for our planned experiments. Finally, we have coordinated a data streaming protocol to allow for DRD motion states to integrate with our system for spatial disorientation detection in real-time. In a recent site visit at NAMRU, we were able to successfully integrate our system with the Langley simulation computer and the DRD computers, streaming data between the three systems, to successfully detect pilot spatial disorientation in real-time during pilot landings on the DRD with a test pilot. Based upon this preliminary data, we have revised our system's parameters, demonstrating technical readiness for our planned

	capstone experiment on the DRD. Our system is a complete implementation to process DRD motion states, continuously simulate our observer model of spatial orientation perception, and compare those model-predicted perceptions to actual Langley lunar landing vehicle states, combined across multiple aspects (linear velocity, vehicle tilt, etc.) and all three axes to yield a continuous, unidimensional metric of spatial disorientation intensity. When the predicted spatial disorientation is too great, it triggers the display countermeasure to help reorient the pilot without requiring unnecessary or excessive burden.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Austin CR, Clark TK. "Spatial orientation perception following a centrifugation-induced gravity transition." 28th European Low Gravity Research Association (ELGRA) Biennial Symposium & General Assembly, Liverpool, UK, September 4-6, 2024. Abstracts. 28th European Low Gravity Research Association (ELGRA) Biennial Symposium & General Assembly, Liverpool, UK, September 4-6, 2024. , Sep-2024
Abstracts for Journals and Proceedings	Folga RV, Etgen CA, HFE N, Clark TK. "Spatial orientation modeling for motion cueing." 2nd International Congress on Aerospace Medicine (ICAM), Lisbon, Portugal, October 3-5, 2024. Abstracts. 2nd International Congress on Aerospace Medicine (ICAM), Lisbon, Portugal, October 3-5, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Clark TK, Rote N, Gopinath AR, Etgen CA, Folga RV, Bailey R, Neuhaus JR, Wood SJ. "Development and validation of motion-based simulation for piloted lunar landing." 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Austin CR, Lonner T, Clark TK. "Tilt perception alterations following sickness induced by centrifugation." 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Rote NC, Gopinath AR, Clark TK. "Spatial orientation modeling choices for assessment of motion-based simulators." 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Lonner TL, Austin CR, Clark TK. "Development of a real-time model of pilot orientation perception and spatial orientation." 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigator's Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Lonner TL, Austin CR, Blake JS, Gupta P, Katz JM, Gopinath AR, Clark T.K. "Changes in tilt perception following sickness induced by centrifugation." The 2025 AsMA-UHMC Annual Scientific Meeting, Atlanta, Georgia, June 1-6, 2025. Abstracts. The 2025 AsMA-UHMC Annual Scientific Meeting, Atlanta, Georgia, June 1-6, 2025. , Jun-2025
Articles in Peer-reviewed Journals	Lonner, TL, Austin CR, Blake JS, Gupta P. Katz, JM, Gopinath AR, Clark TK. "Impact of sickness induced by centrifugation on tilt perception." Front Neurol. 2025 Aug 12;16:1628938. https://doi.org/10.3389/fneur.2025.1628938 ; PubMed PMID: 40881789; PubMed Central PMCID: PMC12380784 , Aug-2025